



PTT PHENOL COMPANY
LIMITED

PROCESS SPECIFICATION

ACID AROMATICS SUMP PUMP (P-2304A)

Docu. No. : 14780-8120-PS-P2304
Project No. : 120117
Rev. No. : Z1
Date : MAY 10, 2016
Page : 1 / 6

TITLE :

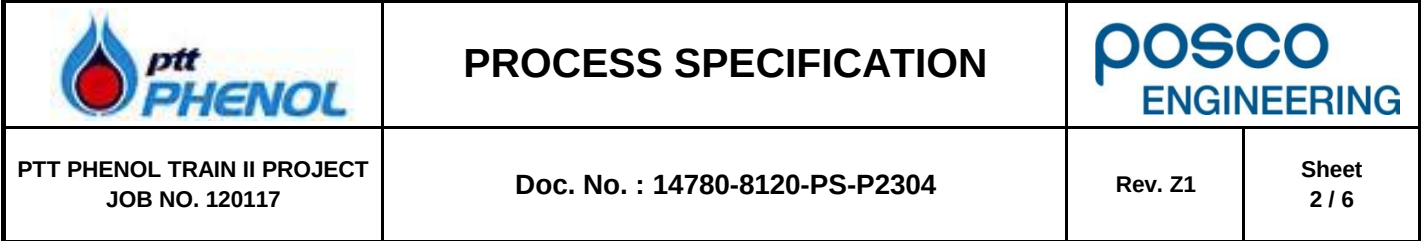
ACID AROMATICS SUMP PUMP

(P-2304A)

DISCIPLINE : **PROCESS**
DOCUMENT NO. : **14780-8120-PS-P2304**
PROJECT NAME : **PTT Phenol Train II Project**
PROJECT NO. : **120117**
OWNER : **PTT Phenol Company Limited (PPCL)**

Z1	MAY 10, 2016	AS BUILT	S.W.Park	J.S.Chung	-	S.W.Lee
F1	MAY 28, 2014	FINAL ISSUE	S.W.Park	J.S.Chung	-	S.W.Lee
A2	FEB 25, 2014	APPROVED FOR DESIGN	S.W.Park	J.S.Chung	-	S.W.Lee
A1	OCT 11, 2013	APPROVED FOR DESIGN	S.W.Park	J.S.Chung	-	S.W.Lee
O1	JUL 30, 2013	COMMENT ISSUE	S.W.Park	J.S.Chung	-	S.W.Lee
REV. NO.	DATE	DESCRIPTION	PREP'N	REVIEW	QA REVIEW	APPROVAL
REV. NO.	PREPARATION DEPT.	PREPARATION	REVIEW	QA REVIEW	APPROVAL	
Z1	PROCESS	S.W.Park	J.S.Chung	-	S.W.Lee	
		MAY 10, 2016	MAY 10, 2016	-	MAY 10, 2016	

This Document is the property of POSCO Engineering Co., Ltd. Therefore, it shall not be released to Any third party without permission of an authorized personnel of the POSCO Engineering Co., Ltd.



Record of Revision	
--------------------	--

[illegible]



UOP LLC | 25 East Algonquin Road | Des Plaines, Illinois 60017-5017 USA

CENTRIFUGAL PUMPS AND DRIVERS

PROJECT SPECIFICATION

963766 - 501

SHEET 3 / 6

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	09-Apr-12	MW	DRL				
1	08-May-12	MW	DRL				
2	13-Aug-12	MW	DRL				

REV													
1	APPLICABLE TO: ☐ PROPOSALS ☐ PURCHASE ☒ AS BUILT												
2	FOR _____ UNIT _____												
3	SITE _____ SERVICE Acid Aromatics Sump Pump												
4	NO. REQ 1 PUMP SIZE _____ TYPE Sump NO. STAGES 1												
5	MANUFACTURER _____ MODEL _____ SERIAL NO. _____												
6	NOTES: INFORMATION BELOW TO BE COMPLETED: ☐ BY PURCHASER ☐ BY MANUFACTURER ☒ BY MANUFACTURER OR PURCHASER												
7	GENERAL												
8	PUMPS TO OPERATE IN (PARALLEL) NO. MOTOR DRIVEN 1 NO. TURBINE DRIVEN _____												
9	(SERIES) WITH _____ PUMP ITEM NO. P-2304A PUMP ITEM NO. _____												
10	GEAR ITEM NO. _____ MOTOR ITEM NO. _____ TURBINE ITEM NO. _____												
11	GEAR PROVIDED BY _____ MOTOR PROVIDED BY _____ TURBINE PROVIDED BY _____												
12	GEAR MOUNTED BY _____ MOTOR MOUNTED BY _____ TURBINE MOUNTED BY _____												
13	GEAR DATA SHT. NO. _____ MOTOR DATA SHT. NO. _____ TURBINE DATA SHT. NO. _____												
14	OPERATING CONDITIONS (5.1.3)												
15	☐ CAPACITY, NORMAL 22.7 m³/h RATED 22.7 m³/h												
16	OTHER _____ m³/h												
17	☐ SUCT PRESS MAX./RATED 3.650 / -0.010000 kg/cm²[g]												
18	☐ DISCHARGE PRESSURE 7.520 kg/cm²[g]												
19	☐ DIFFERENTIAL PRESSURE 7.53 kg/cm²												
20	☐ DIFF. HEAD 81.4 m NPSHA Submerged m												
21	☐ PROCESS VARIATIONS _____												
22	☐ STARTING CONDITIONS _____												
23	SERVICE: ☐ CONT. ☒ INTERMITTENT (STARTS/DAY) _____												
24	☐ PARALLEL OPERATION REQ'D												
25	SITE AND UTILITY DATA												
26	LOCATION: ☐ INDOOR ☐ HEATED ☐ UNDER ROOF												
27	☒ OUTDOOR ☒ UNHEATED ☐ PARTIAL SIDES												
28	☒ GRADE ☐ MEZZANINE _____												
29	ELECTRIC AREA CLASSIFICATION												
30	CL 1 GR IIA DIV ZONE 1												
31	☐ WINTERIZATION REQ'D ☒ TROPICALIZATION REQ'D.												
32	SEE NOTES ON SHEET 6 *												
33	SITE DATA												
34	ALTITUDE _____ BAROMETER _____												
35	RANGE OF AMBIENT TEMPS: MIN/MAX. _____ / _____												
36	RELATIVE HUMIDITY: MIN / MAX _____ / _____ %												
37	UNUSUAL CONDITIONS: ☐ DUST ☐ FUMES												
38	OTHER _____												
39	UTILITY CONDITIONS:												
40	STEAM: DRIVERS HEATING												
41	MIN _____												
42	MAX _____												
43	ELECTRICITY DRIVERS HEATING CONTROL SHUTDOWN												
44	VOLTAGE _____												
45	HERTZ _____												
46	PHASE _____												
47	COOLING WATER:												
48	TEMP. INLET _____ MAX. RETURN _____												
49	PRESS. NORM _____ DESIGN _____												
50	MIN RETURN _____ MAX ALLOW DP _____												
	FOULING FACTOR _____												



UOP LLC | 25 East Algonquin Road | Des Plaines, Illinois 60017-5017 USA

CENTRIFUGAL PUMPS AND DRIVERS

PROJECT SPECIFICATION

963766 - 501

SHEET 4 / 6

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	09-Apr-12	MW	DRL				
1	08-May-12	MW	DRL				

REV	CONSTRUCTION	CONSTRUCTION (CONT)																						
	APPLICABLE STANDARD: ☒ API 610 11TH EDITION ☐ OTHER _____ (SEE REMARKS)	☐ SHAFT DIAMETER BETWEEN BEARINGS _____ mm ☐ SPAN BETWEEN BEARING CENTERS _____ mm ☐ SPAN BETWEEN BEARING & IMPELLER _____ mm	1																					
	PUMP TYPE: <table style="width:100%;"> <tr> <td>☐ OH2</td> <td>☐ BB1</td> <td>☐ VS1</td> <td>☐ VS6</td> </tr> <tr> <td>☐ OH3</td> <td>☐ BB2</td> <td>☐ VS2</td> <td>☐ VS7</td> </tr> <tr> <td>☐ OH4</td> <td>☐ BB3</td> <td>☐ VS3</td> <td>☐ OTHER</td> </tr> <tr> <td>☐ OH5</td> <td>☐ BB4</td> <td>☒ VS4</td> <td></td> </tr> <tr> <td>☐ OH6</td> <td>☐ BB5</td> <td>☐ VS5</td> <td></td> </tr> </table>	☐ OH2	☐ BB1	☐ VS1	☐ VS6	☐ OH3	☐ BB2	☐ VS2	☐ VS7	☐ OH4	☐ BB3	☐ VS3	☐ OTHER	☐ OH5	☐ BB4	☒ VS4		☐ OH6	☐ BB5	☐ VS5		REMARKS: COUPLINGS: DRIVER - PUMP ☐ MAKE _____ ☒ MODEL _____ ☐ RATING _____ ☐ LUBRICATION _____ ☒ LIMITED END FLOAT REQUIRED _____ ☒ SPACER LENGTH _____ mm ☒ SERVICE FACTOR _____	2	
☐ OH2	☐ BB1	☐ VS1	☐ VS6																					
☐ OH3	☐ BB2	☐ VS2	☐ VS7																					
☐ OH4	☐ BB3	☐ VS3	☐ OTHER																					
☐ OH5	☐ BB4	☒ VS4																						
☐ OH6	☐ BB5	☐ VS5																						
	NOZZLE CONDITIONS: <table border="1" style="width:100%; border-collapse: collapse;"> <thead> <tr> <th>SIZE (IN)</th> <th>FLANGE CLASS</th> <th>FAC'G</th> <th>POSITION</th> </tr> </thead> <tbody> <tr> <td>SUCTION</td> <td></td> <td></td> <td></td> </tr> <tr> <td>DISCHARGE</td> <td>300</td> <td>RF</td> <td></td> </tr> <tr> <td>BALANCE DRUM</td> <td></td> <td></td> <td></td> </tr> </tbody> </table>	SIZE (IN)	FLANGE CLASS	FAC'G	POSITION	SUCTION				DISCHARGE	300	RF		BALANCE DRUM				DRIVER HALF COUPLING MOUNTED BY: ☐ PUMP MFR. ☐ DRIVER MFR. ☐ PURCHASER ☐ COUPLING PER API 671	3					
SIZE (IN)	FLANGE CLASS	FAC'G	POSITION																					
SUCTION																								
DISCHARGE	300	RF																						
BALANCE DRUM																								
	PRESSURE CASING CONNECTIONS: <table border="1" style="width:100%; border-collapse: collapse;"> <thead> <tr> <th>NO.</th> <th>SIZE (NPS)</th> <th>TYPE</th> </tr> </thead> <tbody> <tr><td>☐ DRAIN</td><td></td><td></td></tr> <tr><td>☐ VENT</td><td></td><td></td></tr> <tr><td>☐ PRESSURE GAUGE</td><td></td><td></td></tr> <tr><td>☐ TEMP GAUGE</td><td></td><td></td></tr> <tr><td>☐ WARM-UP</td><td></td><td></td></tr> <tr><td>☐ BALANCE / LEAK-OFF</td><td></td><td></td></tr> </tbody> </table>	NO.	SIZE (NPS)	TYPE	☐ DRAIN			☐ VENT			☐ PRESSURE GAUGE			☐ TEMP GAUGE			☐ WARM-UP			☐ BALANCE / LEAK-OFF			BASEPLATES: ☐ API BASEPLATE NUMBER _____ (APPENDIX M) ☐ NON-GROUT CONSTRUCTION:	4
NO.	SIZE (NPS)	TYPE																						
☐ DRAIN																								
☐ VENT																								
☐ PRESSURE GAUGE																								
☐ TEMP GAUGE																								
☐ WARM-UP																								
☐ BALANCE / LEAK-OFF																								
	☐ CYLINDRICAL THREADS REQUIRED	REMARKS: MATERIAL ☐ APPENDIX H CLASS _____ A-8 ☐ MIN DESIGN METAL TEMP _____ 15 °C ☐ BARREL/CASE 316 SS IMPELLER 316 SS ☐ CASE/IMPELLER WEAR RINGS Hard-Faced 316 SS ☐ SHAFT 316 SS ☐ DIFFUSERS _____ ☐ COUPLING SPACER/HUBS _____ ☐ COUPLING DIAPHRAGMS (DISKS) _____	5																					
	CASING MOUNTING: (SEE SEPARATE SHEET FOR VERTICALS) ☐ CENTERLINE ☐ NEAR CENTERLINE ☐ FOOT ☐ SEPARATE MOUNTING PLATE ☒ IN-LINE	BEARINGS AND LUBRICATION BEARING (TYPE/NUMBER): ☐ RADIAL Standard / _____ ☐ THRUST In Driver / _____ ☐ REVIEW AND APPROVE THRUST BEARING SIZE LUBRICATION: ☒ GREASE ☒ FLOOD ☐ RING OIL ☒ FLINGER ☐ PURGE OIL MIST ☐ PURE OIL MIST ☒ CONSTANT LEVEL OILER PREFERENCE (SEE REMARKS) ☐ PRESSURE LUBE SYS ☐ API-610 ☐ API-614 ☒ OIL VISC. ISO GRADE _____	6																					
	CASING SPLIT: ☐ AXIAL ☒ RADIAL		7																					
	CASING TYPE: ☒ SINGLE VOLUTE ☐ MULTIPLE VOLUTE ☐ DIFFUSER ☐ OVERHUNG ☐ BETWEEN BEARINGS ☐ BARREL		8																					
	CASE PRESSURE RATING: ☐ MAX ALLOWABLE WORKING PRESSURE _____ kg/cm ² [g] @ _____ 0 ☐ HYDROTEST PRESSURE _____ kg/cm ² [g] ☐ SUCTION PRESS. REGIONS MUST BE DESIGNED FOR MAWP		9																					
	ROTATION: (VIEWED FROM COUPLING END) ☐ CW ☐ CCW ☐ IMPELLERS INDIVIDUALLY SECURED		10																					
	REMARKS: ☐ BOLT OH3 PUMP TO PAD/FOUNDATION SHAFT: ☐ SHAFT DIAMETER AT COUPLING _____ mm		11																					
	REMARKS: SUCTION TYPE <u>Single</u> IMPELLER TYPE <u>Closed</u> Pumps shall comply with UOP Standard Specifications 5-11 and 7-12 A discharge check valve shall be provided in the bottom of the discharge transfer pipe within the tank to prevent the line from free draining back to the tank when the pump is stopped. This is to ensure the product lubricated bearings do not run dry during each start cycle.		12																					

Note: The information in this document is confidential and the property of UOP LLC and must not be disclosed to others or reproduced in any manner or used for any purpose whatsoever without its written permission.



UOP LLC | 25 East Algonquin Road | Des Plaines, Illinois 60017-5017 USA

CENTRIFUGAL PUMPS AND DRIVERS

PROJECT SPECIFICATION

963766 - 501

SHEET 5 / 6

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	09-Apr-12	MW	DRL				
1	13-Aug-12	MW	DRL				

REV	BEARINGS AND LUBRICATION (CONT)	MECHANICAL SEAL OR PACKING (CONT)	
	☒ OIL HEATER REQ'D ☐ ELECTRIC ☐ STEAM ☐ OIL PRESS TO BE GREATER THAN COOLANT PRESS REMARKS 	☐ VAPOR PRESSURE _____ kg/cm ² [a] @ _____ °C ☐ HAZARDOUS ☐ FLAMMABLE ☐ OTHER _____ ☐ FLOW RATE MAX/MIN _____ / _____ m ³ /h ☐ PRESSURE REQUIRED MAX/MIN _____ / _____ kg/cm ² [g] ☒ TEMPERATURE REQUIRED MAX/MIN _____ / _____ °C QUENCH FLUID: ☐ NAME OF FLUID _____ ☐ FLOW RATE _____ m ³ /h SEAL FLUSH PIPING: ☐ SEAL FLUSH PIPING PLAN ☐ TUBING ☐ CARBON STEEL ☐ PIPE ☐ STAINLESS STEEL ☐ AUXILIARY FLUSH PLAN ☐ API PLAN 74 ☒ TUBING ☐ CARBON STEEL ☐ PIPE ☒ STAINLESS STEEL ☐ PIPING ASSEMBLY: ☒ THREADED ☒ UNIONS ☐ SOCKET WELDED ☐ FLANGED ☒ TUBE TYPE FITTINGS ☒ LOW PRESSURE SWITCH (PLAN 72/74) TYPE _____ ☒ PRESSURE GAUGE (PLAN 72/74) ☒ HIGH FLOW SWITCH (PLAN 72/74) TYPE _____ ☒ FLOW GAUGE (PLAN 72/74) ☐ TEMP INDICATOR (PLANS 21, 22, 23, 32, 41) ☐ HEAT EXCHANGER (PLAN 52/53) REMARKS 	1
	MECHANICAL SEAL OR PACKING SEAL DATA: ☐ SEE ATTACHED API-682 DATA SHEET ☐ NON-API 682 SEAL ☐ APPENDIX H SEAL CODE _____ ☒ SEAL MANUFACTURER _____ ☒ SIZE AND TYPE _____ / _____ Non-contacting Dual ☒ MANUFACTURER CODE _____ SEAL CHAMBER DATA: ☒ TEMPERATURE _____ ☒ PRESSURE _____ kg/cm ² [g] ☒ FLOW _____ m ³ /h ☐ SEAL CHAMBER SIZE (TABLE 2.3) _____ ☐ TOTAL LENGTH _____ mm ☐ CLEAR LENGTH _____ mm SEAL CONSTRUCTION: ☐ SLEEVE MATERIAL 316 Austenitic Stainless Steel ☐ GLAND MATERIAL _____ ☐ AUX SEAL DEVICE _____ ☒ JACKET REQUIRED GLAND TAPS: ☐ FLUSH (F) ☐ DRAIN (D) ☒ BARRIER (B) ☒ QUENCH (Q) ☒ COOLING (C) ☒ LUBRICATION (G) ☒ HEATING (H) ☒ LEAKAGE ☒ PUMPED FLUID (P) ☒ BAL'NCE FLUID (E) ☒ EXTERNAL FLUID INJECTION (X) SEAL FLUIDS REQUIREM'T AND AVAILABLE FLUSH LIQUID: NOTE: IF FLUSH LIQUID IS PUMPAGE LIQUID (AS IN FLUSH PIPING PLANS 11 TO 41), FOLLOWING FLUSH LIQUID DATA IS NOT REQ'D. ☐ SUPPLY TEMPERATURE MAX/MIN _____ / _____ °C ☐ RELATIVE DENSITY (SPECIFIC GRAVITY) _____ @ _____ °C ☐ NAME OF FLUID _____ ☐ SPECIFIC HEAT _____ kcal/(kg.°C) ☐ VAPOR PRESSURE _____ kg/cm ² [a] @ _____ °C ☐ HAZARDOUS ☐ FLAMMABLE ☐ OTHER _____ ☐ FLOW RATE MAX/MIN _____ / _____ m ³ /h ☐ PRESSURE REQUIRED MAX/MIN _____ / _____ kg/cm ² [g] ☐ TEMPERATURE REQUIRED MAX/MIN _____ / _____ BARRIER FLUID: ☐ SUPPLY TEMPERATURE MAX/MIN _____ / _____ °C ☐ RELATIVE DENSITY (SPECIFIC GRAVITY) _____ @ _____ °C ☐ NAME OF FLUID Nitrogen ADDITIONAL REMARKS: 	INSTRUMENTATION VIBRATION: NONCONTACTING (API 670) _____ TRANSDUCER ☐ PROVISION FOR MOUNTING ONLY ☒ FLAT SURFACE REQ'D ☐ SEE ATTACHED API 670 DATA SHEET ☐ MONITORS AND CABLES REMARKS 	2
		STEAM AND COOLING WATER PIPING ☒ COOLING WATER PIPING PLAN _____ ☐ COOLING WATER REQUIREMENTS SEAL JACKET/BRG HSG _____ m ³ /h @ _____ kg/cm ² [g] SEAL HEAT EXCHANGER _____ m ³ /h @ _____ kg/cm ² [g] QUENCH _____ m ³ /h @ _____ kg/cm ² [g] TOTAL COOLING WATER _____ m ³ /h ☐ STEAM PIPING: ☐ TUBING ☐ PIPE REMARKS 	3
			4
			5
			6
			7
			8
			9
			10
			11
			12
			13
			14
			15
			16
			17
			18
			19
			20
			21
			22
			23
			24
			25
			26
			27
			28
			29
			30
			31
			32
			33
			34
			35
			36
			37
			38
			39
			40
			41
			42
			43
			44
			45
			46
			47
			48
			49
			50
			51

Note: The information in this document is confidential and the property of UOP LLC and must not be disclosed to others or reproduced in any manner or used for any purpose whatsoever without its written permission.



UOP LLC | 25 East Algonquin Road | Des Plaines, Illinois 60017-5017 USA

CENTRIFUGAL PUMPS AND DRIVERS

PROJECT SPECIFICATION

963766 - 501

SHEET 6 / 6

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	09-Apr-12	MW	DRL				
1	13-Aug-12	MW	DRL				

REV

NOTES:

* SEE BASIC ENGINEERING DESIGN QUESTIONNAIRE (BEDQ), ENGINEERING DESIGN INFORMATION (EDI) PROJECT 963763 A.4, SECTIONS 3 AND 4.

The following notes apply to those services for which they are specified in this Project Specification:

PUMP NOTES:

2. Pump vendor shall provide non-contacting pressurized dry gas seals. The barrier gas shall be Nitrogen from the plant Nitrogen header available at 7 kg/cm²(g). Pump vendor shall provide a seal control panel for each pump. The panel shall include the following:

Coalescing filter (removal efficiency of 98% on liquids 1 micron or greater and solids 3 microns or greater)
Pressure regulating valve
Pressure gauge
Flow meter low range
Flow meter high range

Alarm switches:

Low Pressure
High Flow

The alarm switches shall activate a control center trouble alarm (by others). The contractor and pump vendor shall coordinate switch selection.

A mechanical amplifier shall be provided by the seal vendor to boost the nitrogen buffer supply pressure when the normal nitrogen supply pressure does not exceed either:

the pump suction pressure by 2.1 kg/cm², or
the pump maximum suction pressure by 0.35 kg/cm² psi.

When these conditions exist, the seal gas control panel scope shall be expanded to include the amplifier and a minimum 2 gallon stainless steel accumulator vessel with local pressure gauge.

6. Mechanical seal parts and packings shall be suitable for handling aromatic hydrocarbons.

9. The "capacity, other" is based on 50% of the lowest normal operating flow.

11. Pumps shall be mounted on cover plates to be furnished by vessel supplier to suit sump tank opening. Each pump shall include a stainless steel suction screen. A steel discharge pipe shall extend through the cover plate terminating in a Cl. 300 flanged connection. Contractor to determine depth of setting. Pump will be spared by common warehouse spare pump P-1304S.

41. The pump shall be designed to be totally interchangeable with phenol train | without modification of existing pump.

42. VOC packing valves should be supplied.

C1. Casing volume to be minimized and subjected to critical volume restriction

C2. Cumene Hydroperoxide is thermally unstable that can initiate self-heating reaction at elevated temperatures

C3. Acid Aromatics sump pump P-2304A shall be identical to the existing Acid Aromatics sump pump in PPCL Unit 1 (Item No. P-1304A/S). Existing warehouse spare pump P-1304S shall be used as a common spare for both services.

MOTOR NOTES:

M5. Non Overloading at all points on pump curve

M10. Motor shall have torque capability to accelerate the pump to operating speed with discharge shutoff valve open, and the discharge check valve closed against a static pressure of 1.8 kg/cm² (g)

M12. Deleted.

M13. The standby pump motor shall be connected to an alternate independent power supply.

GENERAL NOTES:

G1. Lubrication system is applied to high voltage motor's and all pumps' bearings except for vertical and diaphragm pumps.

Note: The information in this document is confidential and the property of UOP LLC and must not be disclosed to others or reproduced in any manner or used for any purpose whatsoever without its written permission.